

Machine Learning and its Applications

Today's goals

Module 1Machine Learning Fundamentals I: Basic Concepts

Module 2Machine Learning Fundamentals II: Model Training and Evaluation

Module 3Data Preparation

Module 9 Introduction

- Introduction to Module 9's Study Video • 2 min
- Why Data Preparation Is Not Optional in Machine Learning Reading • 5 min

Module 9-1 Basic Data Cleaning Review

- Basic Data Cleaning Review Video • 38 sec
- Materials on Basic Data Cleaning Reading • 10 min
- Cleaning Data Is a Decision, Not a Checklist Reading • 5 min

Module 9-2 Distributions, Outliers and Their Removal

- Distributions, Outliers and Their Removal: Gaussian Distribution, Skewness and Outliers Video • 6 min
- Distributions, Outliers and Their Removal: The Z-Score and IQR Method for Outliers Removal Video • 5 min
- Materials on Distributions, Outliers and Their Removal Reading • 10 min
- Outliers, Assumptions, and When “Cleaning” Becomes Damage Reading • 5 min

Module 9-3 Data Transform

- Data Transform: Normalization, Standardization, Power Transform Video • 7 min
- Materials on Data Transform Reading • 10 min
- When Scaling Helps — and When It Quietly Breaks Your Model Reading • 5 min

Module 9-4 Construct Training and Test Sets for Model Evaluation

- Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling Video • 5 min
- Construct Training and Test Sets for Model Evaluation: Cross-Validation Video • 6 min
- Materials on Training/Test Sets Generation Reading • 10 min
- Evaluation Starts with the Split: When Performance Numbers Lie Reading • 5 min

Module 9-5 Data Preparation Overview

- Data Preparation Overview Video • 3 min
- Materials on Data Preparation Reading • 10 min
- From Raw Data to Reliable Learning: What Actually Matters? Reading • 5 min
- Data Preparation and Evaluation Coach Dialogue • 5 min

Module 9 Quiz

- Module 9 Quiz Practice Assignment • 30 min

Module 9 Assignments

- Module 9 Assignment 1 Graded App Item • 30 min
- Module 9 Assignment 2 Graded App Item • 30 min
- Module 9 Assignment 3 Graded App Item • 30 min
- XLS file for Module 9 Assignment Reading • 1 min
- Module 9 Recap Reading • 5 min

Module 4Support Vector Machines

Module 10 Introduction

- Introduction to Module 10's Study Video • 1 min
- Why Support Vector Machines? Reading • 5 min

Module 10-1 Support Vector Machine Fundamentals

- Support Vector Machine Fundamentals: Concepts

Outliers, Assumptions, and When “Cleaning” Becomes Damage

0:00 / 1:24

2x

1. Reflecting on Distribution Assumptions

- Reflect on why the Z-score method for outlier detection depends on data being approximately Gaussian or normally distributed.
- Consider what problems could arise if the Z-score method is applied directly to a heavily skewed dataset without first examining the underlying distribution.

2. Reflecting on Z-Score vs. IQR Methods

- Reflect on the differences between the Z-score and IQR methods for detecting outliers.
- Consider the types of datasets or distributions where each method may be more suitable.
- Reflect on why the IQR method is often considered distribution-agnostic and how this affects its robustness in practice.

3. Reflecting on Whether Outliers Are Always Harmful

- a data collection or measurement error, and
- a genuinely important but rare event.
- Consider how removing each type of outlier could differently affect model training, evaluation, and generalization performance.

4. Reflecting on the Impact of Outlier Removal

- Reflect on how removing outliers may influence statistical properties of the dataset, such as the mean and variance.
- Consider how outlier removal could affect the bias–variance balance of a machine learning model.
- Reflect on whether aggressive outlier removal could unintentionally increase model bias by removing important variability or rare but meaningful patterns from the data.

Mark as completed

 Like  Dislike  Report an issue

Go to next item →